

An Empirical Evaluation of LLMs for Microservice Bad Smell Detection

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Agenda

- ❑ Introduction
- ❑ Methodology
- ❑ Overall Detection Accuracy
- ❑ Detection Performance per Smell
- ❑ Error Analysis
- ❑ Discussion
- ❑ Future Work

Introduction

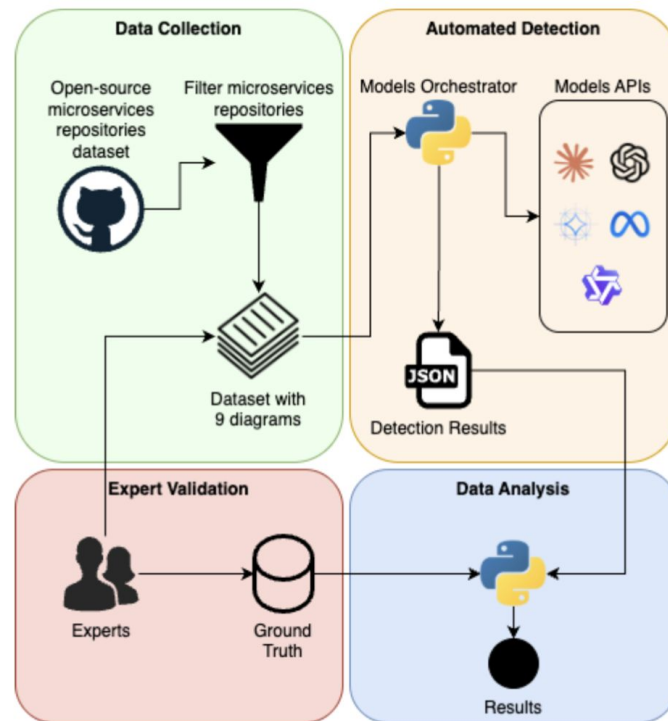
- ❑ Microservices are widely adopted by the industry
- ❑ Systems degrade over time due to poor design decisions
- ❑ Detection is costly and limited
- ❑ Opportunity: Use LLMs

MBS	Description
<i>Cyclic Dependency</i>	Refers to direct or indirect dependency loops between services. It hinders independent deployment and testing, as changes often require coordinated updates across the cycle.
<i>Microservice Greedy</i>	A service that centralizes excessive logic or multiple unrelated domains. It acts as a hub or internal monolith, creating development bottlenecks and single points of failure.
<i>Nano Service</i>	Excessively small services focused on trivial tasks. They lead to high network latency and infrastructure overhead due to fragmented business processes.
<i>No API Gateway</i>	Indicates direct communication between clients and internal services. Bypassing a centralized entry point complicates cross-cutting concerns like authentication and exposes internal structures.
Shared Database	Multiple services accessing the same database schema. This violates the “database-per-service” principle, creating tight coupling and preventing services from evolving their data models independently.

Methodology

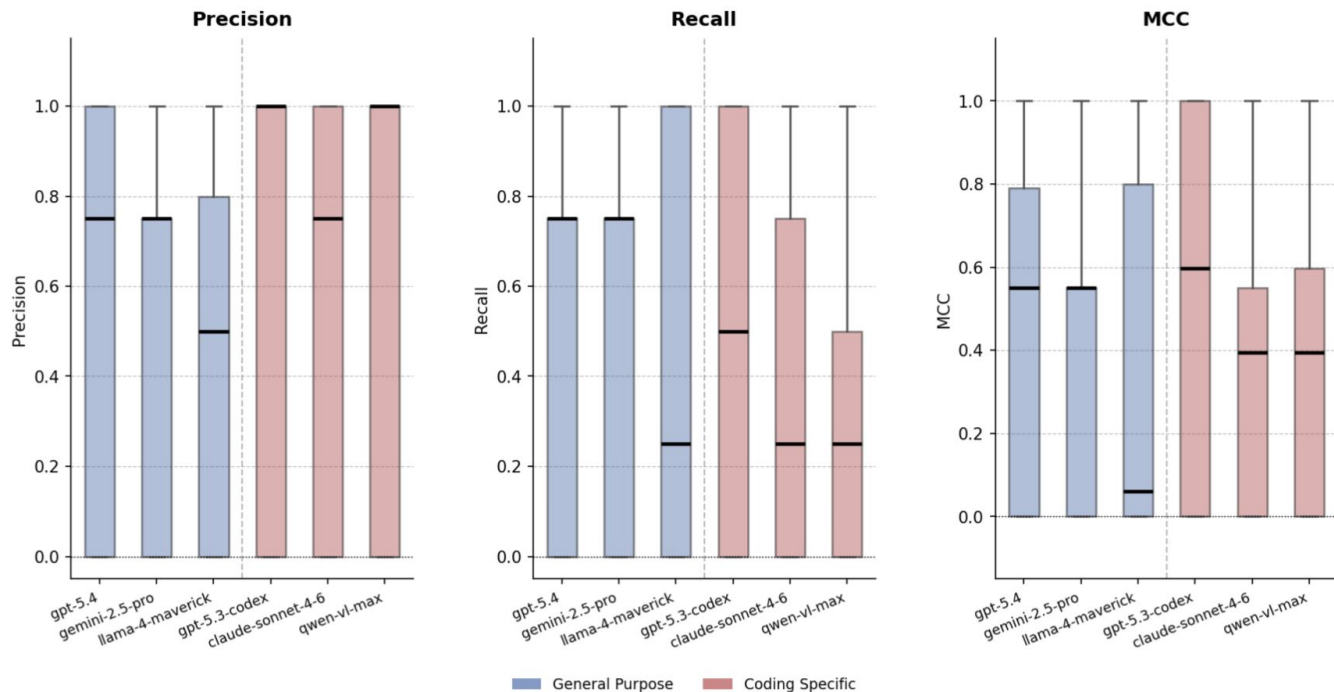
Evaluate 6 LLM models in detecting 5 types of Microservice Bad Smells (MBS) out of 9 Microservice Architectural Diagrams by comparison with a human-validated ground truth

Category	Model	Provider	Open Source
General purpose	GPT-5.4	OpenAI	
	Gemini 2.5 Pro	Google	
	Llama 4 Maverick	Meta	✓
Coding specific	GPT-5.3-Codex	OpenAI	
	Claude 4.6 Sonnet	Anthropic	
	Qwen-vl-max	Alibaba	✓



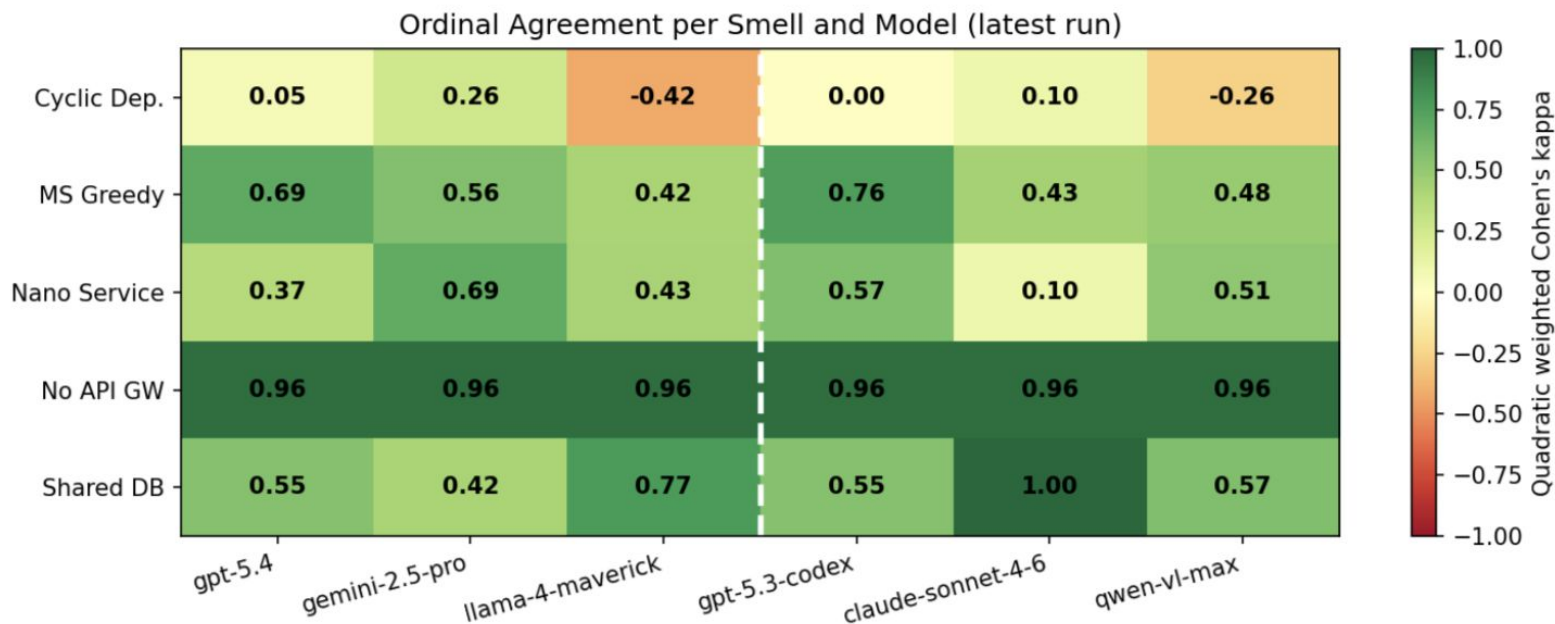
Overall Detection Accuracy

RQ1: To what extent can LLMs accurately Identify MBS from architectural diagrams?



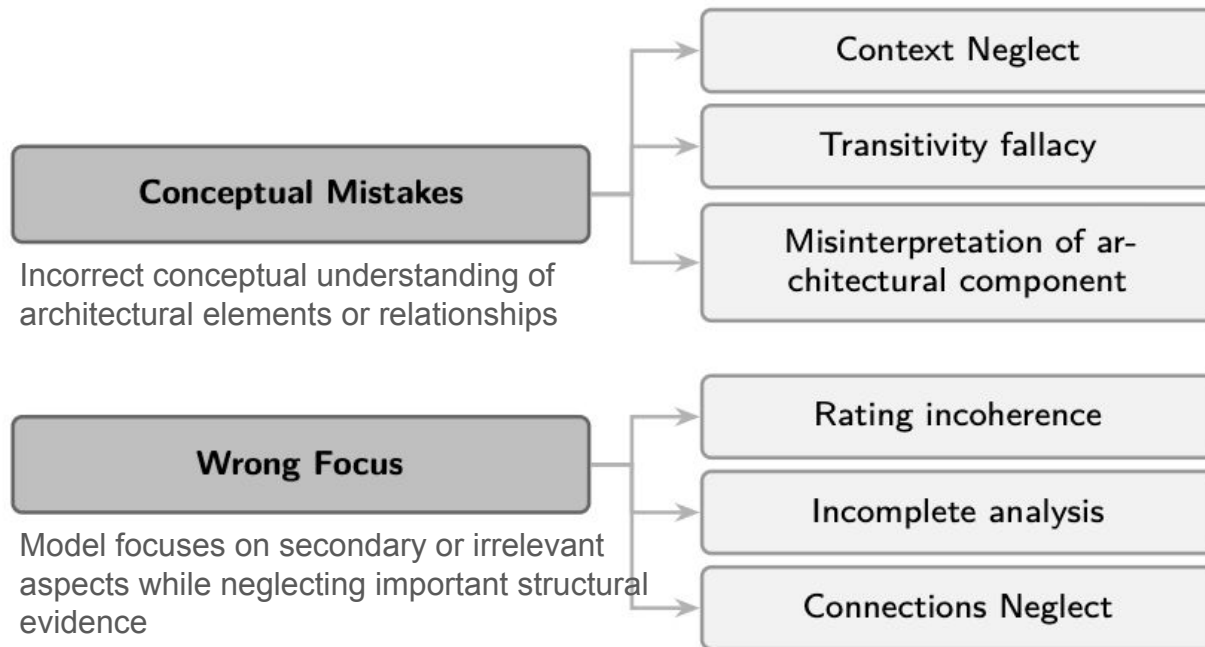
Detection Performance per Smell

RQ2: How does the detection performance of LLMs vary across the selected MBS?



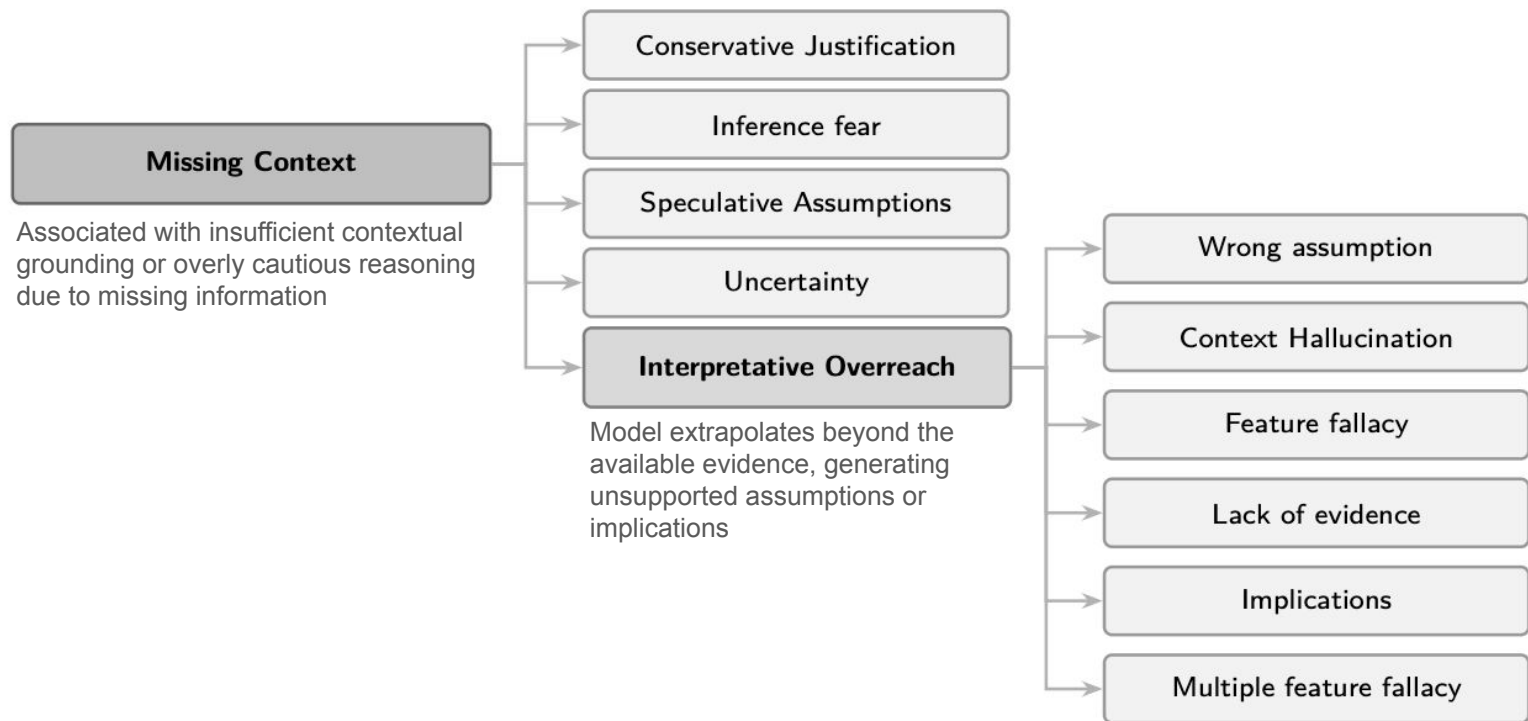
Error Analysis

RQ3: What are the primary root causes of errors made by LLMs during MBS detection?



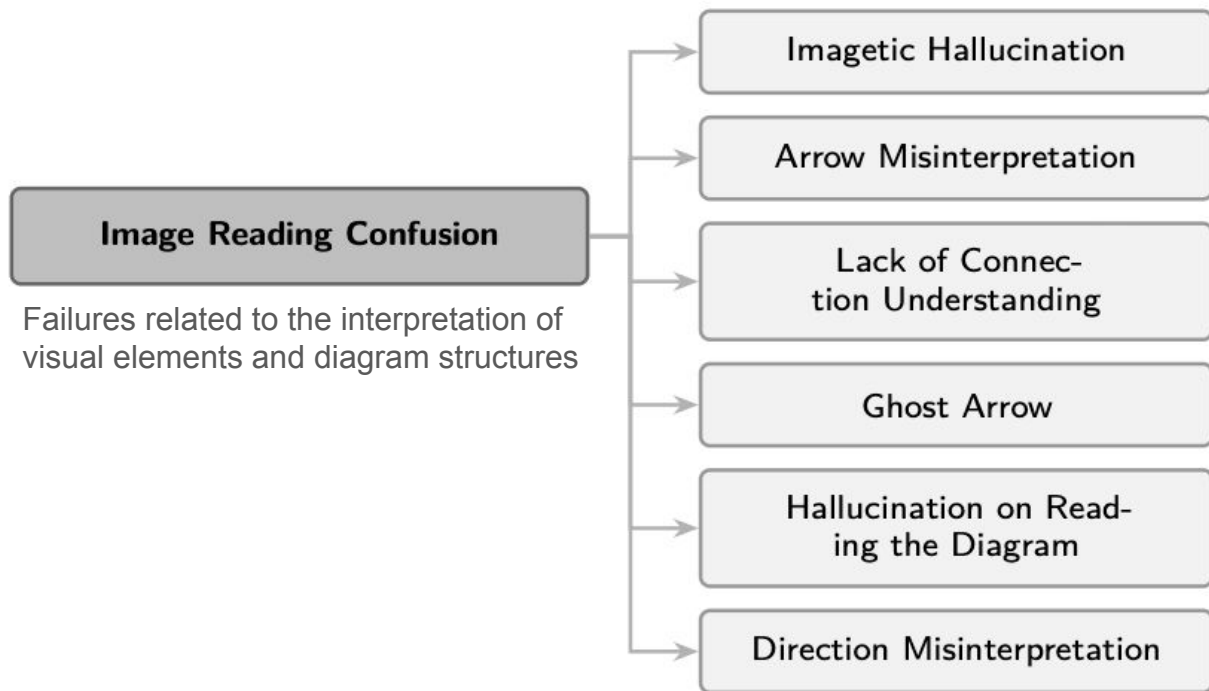
Error Analysis

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Discussion

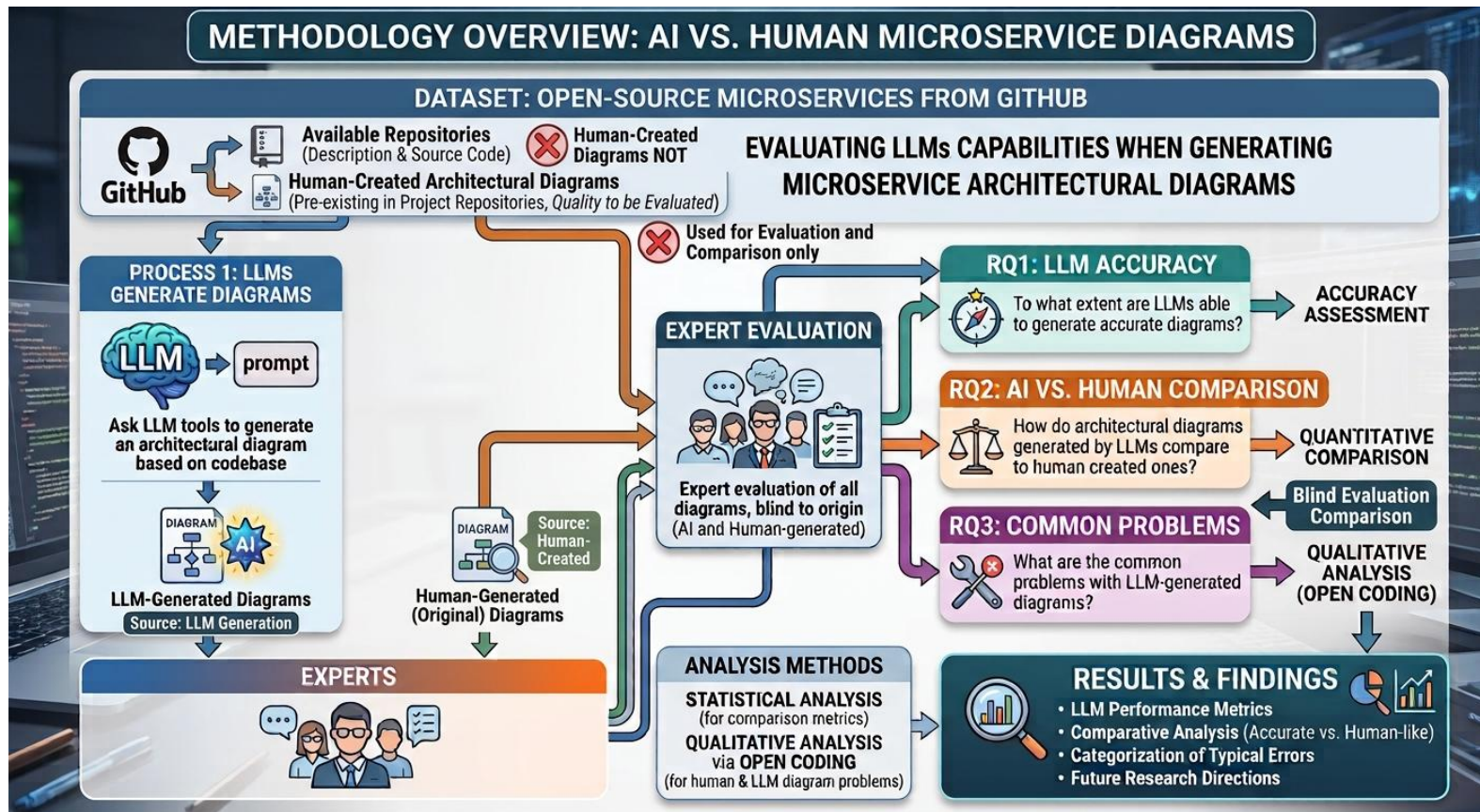
- ❑ **Model specialization matters, but category alone is insufficient:** Coding-specific models achieved highest MCC (0.621) and precision (0.778), yet within-category variance (e.g., 0.144 MCC gap between GPT-5.3-Codex and Claude-Sonnet-4.6) exceeds cross-category averages.
- ❑ **Structural vs. semantic asymmetry:** Structural smells (e.g., missing API Gateway) are detected accurately, but complex patterns (e.g., Cyclic Dependency) suffer false positives; semantic smells yield high precision but low recall due to missing behavioral context.
- ❑ **Practitioner guidance:** Use code-specific multimodal LLMs as assistants with human-in-the-loop; augment diagrams with metadata to reduce hallucinations; prioritize early structural flaw detection.
- ❑ **Research directions:** Move beyond standalone visuals to multi-modal inputs (code, deployment descriptors); develop diagram pre-processing to reduce visual noise; create architecture-specific benchmarks and fine-tuning datasets.

Future Work

Evaluate LLMs Capabilities when Generating Microservice Architectural Diagrams

- ❑ Ask LLMs to generate an architectural diagram based on the available repositories (description and source code) without the diagram
- ❑ Expert evaluation of all the diagrams (both AI and humans generated), then we perform a comparison between the AI vs human generated
- ❑ RQ1: To what extent are LLMs able to generate accurate architectural diagrams?
- ❑ RQ2: How do architectural diagrams generated by LLMs compare to human created ones?
- ❑ RQ3: What are the common problems with LLM-generated diagrams?

Future Work



Thank you!